

Summary and Analysis of Relevant Literatures

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Introduction:

Firstly, I have summarized the literature you provided recently and analyzed its role in our research. Secondly, this document includes a summary of the previous literature available on Moodle, along with the corresponding code (MATLAB and Python) for these literatures. Finally, it presents the models and literature sources of other collected literatures related to "Mathematical Modelling of Mechanical Ventilation using ODEs".

1. Mathematical Medicine and Biology:

Patient-specific modelling of head-up tilt Although this paper **is not** based on the ODE modelling of mechanical ventilation, I still summarized its main content and role:

This paper proposes a patient-specific cardiovascular mathematical model for the head-up tilt test (HUT), aiming to fit the dynamic changes of arterial blood pressure during HUT using heart rate input. The model integrates the effect of lower limb blood pooling caused by gravity and the cardiovascular regulation mechanism. The study screened identifiable parameters through sensitivity analysis, individualized the model by combining nonlinear optimization techniques, and applied it to 5 healthy young subjects. The results show that the model can accurately fit the changes in arterial blood pressure at the carotid bifurcation and can estimate physiological parameters that are not directly measured, all of which fall within the physiologically reasonable range. Model verification indicates that its prediction results are consistent with clinical observations, supporting the dominant role of the carotid sinus baroreceptor in blood pressure

regulation during HUT.

Our project aims to integrate the ODE models of the respiratory and cardiovascular systems, analyze the impact of key parameters on cardiopulmonary variables, and verify the model to support clinical applications. This literature can provide methods for cardiovascular system modelling, ideas for parameter estimation and model verification, and verify the predictive ability of the integrated model for the impact of mechanical ventilation (e.g., PEEP). It can also provide inspiration for analyzing the "mechanical interaction between respiration and cardiovascular system", especially having reference value for the mathematical expression of the cardiovascular part and the analysis of key parameters.

2. Hann CMPB 2013

This paper aims to verify a subject-specific mathematical model of the cardiovascular system (CVS). By adopting an iterative proportional gain identification method, the model overcomes the limitation of previous studies that relied on complete experimental data.

To reduce the number of parameters identified simultaneously and improve convergence stability, the model simplifies the systemic circulation and pulmonary circulation into sub-models and identifies them step by step. The study verified the model on an acute pulmonary embolism (APE) model of 5 pigs: pathological states were induced by inserting autologous thrombi, and relevant pressure and volume waveforms were continuously measured. The results show that the matching error of the model for the input data used for identification (e.g., P_{ao} , P_{pa}) is $< 0.5\%$; the median absolute errors of the model's predictions for the average ventricular volume and maximum pressure (which were not used for fitting) are 4.3% and 4.4% respectively, which is comparable to the error range ($5\%-10\%$) of existing monitoring equipment in the ICU. This confirms the application potential of the model in critical care.

This paper provides specific methods for ODE modelling of the cardiovascular system, parameter identification strategies,

verification ideas for key variables, and modelling references for cardiopulmonary mechanical interaction for the project, which is of great reference significance for the integration of respiratory-cardiovascular models.

3. Jermey Physiol.

Genomics 2011 The core of this literature is to construct a semi-mechanistic mathematical model, which reveals the regulatory mechanism of the autonomic nervous system on cardiac function and heart rate variability (HRV) in human endotoxemia (lipopolysaccharide, LPS injection to simulate systemic inflammation). It fills the research gap in the "association between cellular molecular mechanisms and HRV changes".

This literature **is irrelevant to** the ODE modelling of mechanical ventilation. The ODEs in this literature are only used to describe the dynamics of "inflammatory factors, hormones, and autonomic nerve activity", not to simulate the impact of "mechanical ventilation on intrathoracic pressure, venous return, and cardiopulmonary interaction".

The literature provides key references for "ODE modelling methods of the cardiovascular system, autonomic nerve-cardiac regulation mechanisms, and model verification logic", but does not involve the respiratory system or mechanical ventilation. As a classic study on "multi-scale cardiovascular ODE modelling", it provides key cardiovascular modelling paradigms, autonomic nerve regulation mechanisms, verification logic, and clinical transformation ideas for the user's "respiratory-cardiovascular integrated model". It can significantly reduce the difficulty for users to construct the cardiovascular module and improve the mechanistic nature and clinical applicability of the model.

4. Khoo

This literature (Khoo, 1990) focuses on the model-based evaluation of the single-breath CO₂ ventilatory response test (SB CO₂ test). Its core is to simulate and verify the accuracy of quantifying the

peripheral chemoreflex gain (G_p) in this test through a reciprocal breathing model.

This paper uses ODE modelling, but does not directly focus on mechanical ventilation modelling. The ODEs of the respiratory module in this literature can be directly referenced: Dynamic changes of alveolar CO_2 (inspiratory phase):

$$(V_L + V_t)\dot{P}_{ACO_2} = \dot{V}_t(P_{D_{CO_2}}^5 - P_{ACO_2}) + 863\dot{Q}(C_{\bar{v}CO_2} - C_{aCO_2}) \quad (A1)$$

Cardiovascular mixed convection effect:

$$T_1 T_2 \ddot{P}_{aCO_2} + (T_1 + T_2) \dot{P}_{aCO_2} + P_{aCO_2} = P_{ecCO_2}(t - T_d)$$

5.R. Rodriguez-Roisin 1994

This literature presents a clinical study on patients with mechanical ventilation for Chronic Airflow Obstruction (CAO) complicated by Acute Respiratory Failure (ARF). Its core focus is exploring the effects of extrinsic Positive End-Expiratory Pressure (PEEP) and intrinsic PEEP (PEEPi) on ventilation/perfusion (V/Q) imbalance, respiratory mechanics, and hemodynamics.

The study obtains empirical data on the actual effects of PEEP on cardiopulmonary function through human experiments, and does not involve any mathematical modeling. This literature provides clinical verification criteria for "PEEP-related effects" for the ODE model.

5. Smith 2010

The core of this literature is the development and verification of a mechanical ventilation Decision Support System (DSS) based on physiological models and decision theory, which is designed to optimize mechanical ventilation settings for ICU patients.

This literature is based on ODE modeling for mechanical ventilation:

Dynamic physiological processes (such as gas transport and

acid-base balance) must be constructed using ODEs: - ODEs are used to describe "the rate of change of oxygen concentration over time" (e.g., $d[O_2]/dt = \text{ventilation intake rate} - \text{tissue consumption rate}$); - ODEs are also applied to depict the transport rate of CO_2 between different compartments (blood, interstitial fluid, and tissues).

For example: The formula describing the relationship between tidal volume (V_T), peak inspiratory pressure (PIP), and Positive End-Expiratory Pressure (PEEP) is the core equation of the pulmonary mechanics model, which is used to predict changes in PIP under

volume-controlled ventilation:
$$C = \frac{V_T}{PIP - PEEP}$$

Calculation of oxygen consumption:

$$VO_2 = Q \times (CaO_2 - CmvO_2)$$

Calculation of blood oxygen content:

$$CxO_2 = (\alpha_{O_2} \times PxO_2) + (eHb \times SxO_2)$$

Previous literature

1. Burkhoff 1993

Burkhoff and Tyberg (1993) developed a comprehensive theoretical model to explore the hemodynamic mechanisms underlying the elevation of pulmonary venous pressure (PVP) following acute left ventricular dysfunction (LVD). The model conceptualizes the cardiovascular system using lumped parameters, with ventricles represented by time-varying elastance $E(t)$, defined as

$P(t) = E(t)[V(t) - V_0]$, where $P(t)$ is ventricular pressure, $V(t)$ is ventricular volume, and V_0 is the volume at zero pressure. Systolic elastance follows a raised sine wave, while diastolic elastance decays exponentially with a time constant τ . The pulmonary and systemic vasculatures are modeled as networks of resistive (R) and capacitive (C) elements, with vascular pressure related to stressed volume V_S by $P = V_S/C$, where $V_S = V_{\text{total}} - V_U$ (V_U is unstressed volume). Key simulations involve reducing left ventricular end-systolic elastance (E_{es}) by 50%, demonstrating that PVP rises minimally with LVD alone. However, when V_U decreases (mimicking sympathetic activation), PVP increases significantly due to a shift of blood from the unstressed to the stressed compartment, increasing V_S . Pericardial constraints are incorporated via a nonlinear function of combined ventricular volumes: $P_{\text{pericard}} = a \exp[\beta(V_{LV} + V_{RV})]$, which modulates filling pressures and limits cardiac output responses. This model quantifies how PVP elevation is driven by venous volume redistribution rather than direct LVD effects, emphasizing V_U as a critical variable in pathophysiology.

2. Olufsen MB 12

Olufsen et al. (2013) presented a coupled cardiovascular-respiratory lumped-parameter model to predict patient-specific responses to hypercapnia in congestive heart failure (CHF). The cardiovascular component includes 3 arterial, 3 venous, and 2 ventricular compartments, with pressure dynamics in each vascular compartment governed by $dp_i(t)/dt = q_{in}(t) - q_{out}(t)/C_i$, where q_{in} and q_{out} are inflow and outflow, and C_i is compliance. Ventricles are modeled using time-varying elastance $E(t)$, defined piecewise:

$$E(t) = \begin{cases} (E_S - E_D) \left[1 - \cos\left(\frac{\pi t}{T_M}\right) \right] / 2 + E_D & 0 \leq t \leq T_M \\ (E_D - E_S) \left[\cos\left(\frac{\pi(t-T_M)}{T_R}\right) + 1 \right] / 2 + E_D & T_M \leq t \leq T_M + T_R \\ E_D & \text{otherwise} \end{cases}$$

Here, E_S (end-systolic) and E_D (end-diastolic) are elastances, while T_M (time to maximum elastance) and T_R (relaxation time) are fractions of the cardiac cycle. Cerebral vascular resistance RB is

captured via piecewise linear splines: $R_B(t) = \sum_{i=1}^n \gamma_i H_i(t)$, with $H_i(t)$ as hat functions, to model CO₂-induced vasodilation. The respiratory model includes systemic and cerebral tissue compartments, with gas exchange equations: tissue $\bar{V}_{T,g} \frac{dc_{T,g}}{dt} = M_{T,g} - D_{T,g}(c_{T,g} - c_{S,g})$ (metabolism M , diffusion D) and capillaries $\bar{V}_{T,g} \frac{dc_{T,g}}{dt} = M_{T,g} - D_{T,g}(c_{T,g} - c_{S,g})$. Parameter estimation via the Levenberg-Marquardt method optimizes fits to clinical data (arterial pressure, cerebral blood flow velocity), enabling accurate prediction of cerebral CO₂ reactivity

3. PEEP math 2011

Chiew et al. (2011) developed a mathematical model to optimize positive end-expiratory pressure (PEEP) in mechanical ventilation for ARDS/ALI patients, utilizing a single-compartment linear lung model. Airway pressure is described

by $P_{aw}(t) = E_{drs}(t)V(t) + R_{lung}Q(t) + P_0$, where $E_{drs}(t)$

(dynamic elastance) is time-variant, $V(t)$ is volume, $Q(t)$ is flow, R_{lung} is resistance, and P_0 is offset pressure. Constant lung elastance E_{lung} and dynamic $E_{drs}(t)$ are identified using integral-based methods, with median fitting errors of 0.9% for $E_{drs}(t)$ and 5.6% for E_{lung} . Three key metrics guide PEEP optimization: 1) Minimum E_{lung} or E_{drs} , indicating maximal lung compliance; 2) E_{drs} Area, calculated as the integral of

$E_{drs}(t)$ over inspiration ($\int E_{drs}(t)dt$), which correlates strongly with

work of breathing ($WOB_E = E_{lung}V^2$); 3) Inflection points in E_{drs} -PEEP curves (5-10% above the minimum E_{drs}), representing diminishing returns in recruitment. Correlations confirm E_{drs} Area relates to WOB_E ($R = 0.936$) and median E_{drs} ($R = 0.896$), validating its use in personalized PEEP selection to balance recruitment and avoid overdistension.

4. Olufsen

Olufsen et al. (2011) extended a cardio-respiratory model to investigate responses to inhaled CO_2 in 1CHF patients, focusing on end-tidal CO_2 and cerebral blood flow velocity (CBFV). The respiratory model includes an alveolar compartment and three dead space compartments ($D1, D2, D3$), with alveolar gas partial pressure dynamics:

$$V_A \frac{dp_{A,g}}{dt} = \begin{cases} \dot{V}_{IE}(p_{D3,g} - p_{A,g}) + 0.98 \cdot 863 q_P (c_{v,g} - c_{a,g}) & \text{inspiration} \\ 0.98 \cdot 863 q_P (c_{v,g} - c_{a,g}) & \text{expiration} \end{cases}$$

Here, V_A is alveolar volume, $\dot{V}_{IE}$ is airflow, $p_{D3,g}$ is pressure in the third dead space, q_P is pulmonary

flow, and $c_{v,g}/c_{a,g}$ are venous/arterial gas concentrations. Dead space pressure changes follow:

inspiration $V_{D_i} \frac{dp_{D_i,g}}{dt} = \dot{V}_{IE}(p_{D_{i-1},g} - p_{D_i,g})$ ($D1$ uses inspired gas pressure $p_{I,g}$); expiration

$$V_{D_i} \frac{dp_{D_i,g}}{dt} = \dot{V}_{IE}(p_{D_i,g} - p_{D_{i+1},g}) \quad (D3 \text{ connects to alveoli}).$$

Tissue-capillary gas exchange mirrors Olufsen et al. (2013), with capillary compartments critical for accurate end-tidal CO_2 dynamics. CBFV increases due to CO_2 -induced reductions in cerebral vascular resistance RB , modeled via piecewise linear splines, enabling quantification of cerebral CO_2 reactivity in CHF.

5. marino math

Marini et al. (1989) derived mathematical formulas for pressure-preset ventilation (PPV) modes, describing key variables such as tidal volume (V_T), minute ventilation (V_E), auto-PEEP, and work of breathing. Tidal volume follows exponential dynamics:

inspiration $V_i = (V_f - V_{EE})(1 - e^{-k_I t})$, where $V_f = P_{\text{set}} \cdot C$ (equilibrium volume), V_{EE} is end-expiratory volume, and $k_I = 1/(R_I C)$ (R_I inspiratory resistance);

expiration $V_e = (V_T + V_{EE})e^{-k_E t}$, where $k_E = 1/(R_E C)$ (R_E

expiratory resistance). Minute ventilation is $V_E = f \cdot V_T$, with a theoretical maximum:

$$\dot{V}_{E,\max} = \frac{60P_{\text{set}}D(1-D)}{R_ED + R_I(1-D)}$$

where $D = T_I/T_T$ (duty cycle, T_I inspiratory time, T_T total cycle time).

Auto-PEEP (PEE) is calculated as $PEE = \frac{V_T e^{-k_E T_E}}{C(1-e^{-k_E T_E})}$, and mean

alveolar pressure P_m integrates inspiratory and expiratory

volume-time areas: $P_m = \frac{A_{\text{in}} + A_{\text{de}}}{T_T C}$. Work per minute, a measure of ventilator power, is quadratic in P_{set} :

$$\dot{W}_m = \frac{6P_{\text{set}}^2 D(1-D)}{R_ED + R_I(1-D)}$$

These formulas highlight resistance and duty cycle as primary determinants, providing a framework to optimize PPV settings in clinical practice.

The literature we have found:

Ordinary-Differential-Equation (ODE) Models for Patient-Specific
Optimisation of Mechanical Ventilation: A Mini-Review and
Method Synthesis

Mechanistic ODE models remain the workhorse for bedside respiratory mechanics because they balance physiological interpretability and identifiability. This mini-review synthesises five

ODE-based studies spanning single-compartment lung mechanics, recruitment/derecruitment (RD) dynamics, time-varying elastance for spontaneous breathing, and parameter-updating strategies for clinical optimisation of positive end-expiratory pressure (PEEP). Collectively, these works show that (i) simple linear ODEs fitted to airway pressure–flow–volume signals can robustly estimate resistance (R) and elastance (E) at the bedside; (ii) adding RD dynamics via low-order ODEs or state models enables PEEP titration grounded in physiology; and (iii) time-varying elastance and data-driven updating improve applicability to partially assisted breathing and low-PEEP regimes. We summarise equations, identification methods, and clinical workflows, compare strengths/limitations, and outline a practical protocol that integrates these ideas for independent group projects.

Mechanical ventilation (MV) management hinges on selecting protective settings tailored to patient-specific lung mechanics, particularly PEEP. ODE-based models have been repeatedly used at the bedside because they provide interpretable parameters (compliance/elasticity and resistance), can be identified from routine waveforms, and are computationally light for real-time decision support.

Single-compartment equation of motion (controlled MV)

$P_{aw}(t) = R \cdot Q(t) + E \cdot V(t) + P_0$, where P_{aw} is airway pressure, Q =flow, V =volume, R airway resistance, E elastance, and P_0 PEEP. Identifying these parameters from data enables bedside mechanics estimation.

Recruitment–derecruitment (RD) dynamics

ODEs describe fraction of open lung units evolving with pressure and time, underpinning model-based PEEP strategies.

Time-varying elastance

For spontaneous breathing, define time-varying elastance $E_{drs}(t)$ to capture demand and chest-wall effects.

Parameter updating at low PEEP

Introduce updating and hysteresis-loop descriptors to sustain identifiability and prediction accuracy.

Model Type	Core Feature	ODE	Clinical Application	Strengths	Limitations
Single-Compartment	Constant R, E		Fully controlled	Simple, fast computation,	Masks regional heterogeneity; not suitable for spontaneous breathing

RD-Integrated	Time-dependent $\phi(t)$	ventilation (sedated patients) ARDS patients (PEEP optimization)	parameters easy to interpret Quantifies alveolar recruitment; reduces VILI risk	Requires dynamic PEEP maneuvers; P_{open}/P_{close} non-unique
Time-Varying Elastance	$E_{drs}(t) = E_{cw}(t) + E_{demand}(t)$	Partially assisted ventilation (weaning)	Captures patient effort; non-invasive	Relies on high-quality flow/volume signals; sensitive to motion artifacts
Parameter-Updating	RLS + hysteresis descriptors	Low-PEEP regimes (weaning, barotrauma risk)	Stable estimates; adapts to signal noise	Increases computational load; requires P-V loop analysis

Single-compartment ODEs mask heterogeneity, RD parameters may be non-unique. Nevertheless, they are practical, interpretable, and clinically implementable.

ODE-based lung models balance interpretability, identifiability, and clinical practicality. Combining equation-of-motion fitting, RD-based PEEP metrics, time-varying elastance, and parameter updating provides a defensible pipeline for clinical optimisation and research projects.

Reference

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